Soumya Roy

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Research Profile

I am a machine learning researcher with experience in interpretable and explainable AI, applied to high-stakes prediction tasks across clinical risk and behavioural health. I completed my MSc in Advanced Computer Science at the University of Liverpool. Across these projects, I have used techniques to extract reliable signal from incomplete and imbalanced tabular datasets. I am now seeking doctoral research at the intersection of tabular machine learning and large language models, with the goal of building methods that are both rigorously grounded and practically deployable for real-world structured data, such as statistical report generation and semantic search.

Education

- **University of Liverpool** — *September 2024 – September 2025*
  MSc Advanced Computer Science — Liverpool, United Kingdom
  - **Award:** Merit
  - **Relevant modules:** Efficient Algorithm Design and Analysis, Computational Intelligence, Geographic Data Science, Safety and Dependability in Software Systems
  - **Thesis:** *A Web Travel Service* – Created a full-stack web platform connecting tourists with local service providers, featuring multi-role authentication, booking management, and interactive map integration using React.js, Node.js, and MongoDB.
- **Indian Institute of Engineering Science and Technology** — *August 2019 – August 2023*
  Bachelor of Technology in Information Technology — Howrah, India
  - **Grade:** 8.4/10
  - **Relevant modules:** Machine Learning, Discrete Mathematics and Graph Theory, Database Management Systems, Object-Oriented Programming, Computer Organisation and Architecture

Publications & Preprints

- Roy, S. and Sarkar, S.D. (2026). *Explainable Early Prediction of Acute Hypotensive Episodes in ICU Patients Using Routine Vital Signs: An eICU Study.* Zenodo. https://doi.org/10.5281/zenodo.19161469. *[Preprint]*
- Sarkar, S.D. and **Roy, S.** (2026). *Machine Learning-Based Risk Factor Analysis for Cervical Cancer Prediction: A Comparative Study of Class Imbalance Handling Strategies with SHAP-Driven Clinical Risk Stratification.* Zenodo. https://doi.org/10.5281/zenodo.19368694. *[Preprint]*
- Roy, S. and Sarkar, S.D. (2026). *Local Bounded Stochastic Imputation for Ordered Numerical Sequences.* Zenodo. https://doi.org/10.5281/zenodo.19429162. *[Preprint]*
- Sarkar, S.D. and **Roy, S.** (2026). *A Mathematical Index for Quantifying and Reducing Student Depression Risk Through Personalised Behavioural Guidance.* Zenodo. https://doi.org/10.5281/zenodo.19457072. *[Preprint]*

Research & Professional Experience

- **Indian Institute of Engineering Science and Technology** — *May 2022 – July 2022*
  Research Intern – Solar Power Forecasting — Howrah, India
  - Built and compared deep learning models (MLP, RNN, LSTM) to forecast short-term solar power output using weather and sensor data from 21 photovoltaic sites across Germany.
  - Used sliding window approaches and feature extraction techniques (PCA, 1D CNN, Autoencoders) to prepare and condense large time-series datasets for model input.
  - Compared deep learning models against standard ML baselines, finding consistent accuracy gains for renewable energy forecasting tasks.
- **Avanseus Technologies Pvt Ltd** — *February 2024 – June 2024*
  Software Engineer Intern — Bengaluru, India
  - Developed a full-stack Employee Access Portal using React and Spring Boot, designing RESTful APIs that reduced authentication response time by 40%.
  - Improved system efficiency by 30% through database query optimisation and caching techniques.

Teaching Experience

- **Dixons Fazakerley Academy** — *November 2025 – Present*
  Teaching Assistant — Liverpool, United Kingdom
  - Delivered lessons and supported individual students, adjusting my approach based on each student's needs.
- **Hillside High School** — *September 2025 – November 2025*
  Teaching Assistant — Liverpool, United Kingdom
  - Helped deliver lessons and worked with students one to one across different subject classes.

Technical Skills

- **Programming Languages:** Python, C++
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **NLP & LLMs:** LangChain, RAG, Hugging Face Transformers
- **Biomedical & Image Analysis:** MONAI, OpenCV
- **Web Technologies:** React, Node.js, Express, HTML, CSS, JavaScript
- **Database Systems:** MySQL, MongoDB
- **Tools & Platforms:** Git, GitHub, Docker, Power BI, Postman

Certifications

- **Advanced Learning Algorithms** — *March 2025*
- **Python for Data Science, AI & Development** — *April 2025*
- **Understanding and Visualizing Data with Python** — *June 2025*

Awards & Achievements

- **Merit Scholarship —** Awarded for academic excellence on admission to MSc programme. — *2024*
- **Institute Rank 1 —** Achieved in problem solving on the GeeksforGeeks coding platform — *2024*

Referees

Prof. Dominik Wojtczak
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